



Hostel Security App

AI-Powered Hostel Security Monitoring System

Real-time YOLOv8 detection · Live dashboard · Smart security chatbot

Hackathon Project Presentation

The Problem

Hostels struggle with round-the-clock manual surveillance



No Continuous Monitoring

Wardens can't watch every camera feed 24/7 — incidents go unnoticed.



Unauthorized Entry

Intruders and non-residents enter premises without detection.



Restricted Area Breach

Entry into staff-only or restricted zones often goes unchecked.



Overcrowding Risk

Common areas can exceed safe occupancy without any alert.

Project Goal

Develop a smart security system that monitors hostel premises using live CCTV/webcam footage, detects suspicious activities using AI, generates real-time alerts, and gives wardens a dashboard with an AI security chatbot.

1 Detect (YOLOv8)

2 Alert & Store

3 Dashboard View

4 AI Chat Q&A

4

Detection Types

24/7

Live Monitoring

1

AI Assistant

Technology Stack

An end-to-end pipeline from live video to natural-language answers



AI Module

Python, OpenCV, YOLOv8



Backend

Python — Flask / FastAPI



Database

MongoDB / Firebase / SQLite



Frontend

React.js + Tailwind CSS

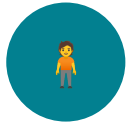


Chatbot

OpenAI API / Gemini API

AI Detection Module

What YOLOv8 + OpenCV watches for on every video frame



Person Detection

Real-time detection & live people count on every frame



Intruder Entry

Virtual entry-line trips an alert on unauthorized crossing



Restricted Zones

User-defined areas flag entry the moment they're breached



Loitering

Flags anyone lingering in a restricted zone too long



Overcrowding

Alerts once people count passes a safe threshold

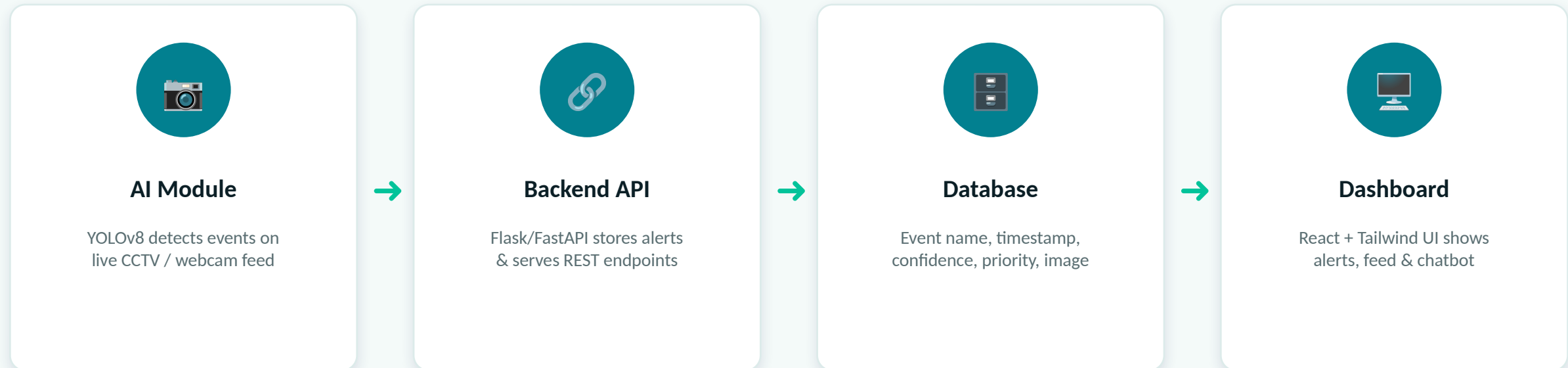


Evidence Logging

Saves snapshot + event, time, confidence & priority to DB

System Architecture

How detection flows from the camera to the warden's screen



Backend REST APIs expose: current alerts · alert history · evidence images · live people count · security status · warden authentication

Dashboard, Secure Login & AI Chatbot

Everything a warden needs, in one screen



Admin / Warden Login

- Username & password authentication
- Role-based access for wardens/admins
- Protects the security dashboard

- Live CCTV monitoring window + real-time people count
- Instant alert notifications with priority levels
- Searchable, filterable alert history table + evidence viewer
- Sound + browser notifications for critical events



Hostel Security Assistant

"Was there any intruder detection?"

"Yes, an intruder alert was detected at 10:30 PM."

Floating assistant, always available bottom-right of the app

Conclusion & Future Scope

Hostel Security App turns any webcam into a 24/7 intelligent security guard for hostel wardens.



Mobile App

Push alerts straight to warden's phone



Multi-Camera

Scale detection across every hostel block



Analytics Reports

Weekly/monthly automated security summaries



Face Recognition

Identify known residents vs. strangers

Thank You